

Database Design

AI Innovation Observatory for Central Banking & Financial Sector Intelligence

Bank of Tanzania — Student Project

1. Overview

Property	Value
Engine	SQLite 3 (file-based)
ORM	SQLAlchemy 2.0
Database file	backend/observatory.db
Schema definition	backend/app/models.py
Tables	4 (publications, sources, alert_rules, rule_matches)
Upgrade path	Change DATABASE_URL in .env to PostgreSQL — no code changes

2. Entity Relationship Diagram

SOURCES		
int	id	PK
string	name	UK
string	url	
string	method	
boolean	active	
boolean	builtin	
string	added_by	
datetime	added_at	



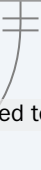
 produces (via collector)

PUBLICATIONS		
int	id	PK
string	title	
string	institution	
string	source_url	
datetime	published_date	
string	document_type	
text	abstract	
text	raw_content	
string	fingerprint	UK
text	ai_summary	
string	ai_topics	
text	ai_relevance	
boolean	ai_processed	
datetime	ai_processed_at	
string	ai_engine	
boolean	hidden	
boolean	manual	
datetime	collected_at	

ALERT_RULES		
int	id	PK
string	keyword	
string	source	
string	topic	
string	priority	
string	created_by	
datetime	created_at	
boolean	active	



 matched by



 matched to

RULE_MATCHES		
int	id	PK
int	rule_id	FK
int	publication_id	FK
datetime	matched_at	

Relationships: There are no strict foreign key constraints — the schema uses logical references (dotted line) instead of enforced FKs. This keeps SQLite simple and lets rows survive if a source or rule is deleted. Orphaned rows are surfaced as warnings in the UI.

3. Table: `publications`

The core table — every collected or manually added document.

Column	Type	Nullable	Purpose
<code>id</code>	Integer	No	Primary key
<code>title</code>	String	No	Publication title
<code>institution</code>	String (indexed)	No	BIS / IMF / World Bank / CBK / custom
<code>source_url</code>	String	No	Link to original — never overwritten (traceability)
<code>published_date</code>	DateTime (indexed)	Yes	When the source published it
<code>document_type</code>	String	Yes	<code>press_release</code> / <code>speech</code> / <code>working_paper</code> / ...
<code>abstract</code>	Text	Yes	Short summary from the source
<code>raw_content</code>	Text	Yes	Full text (when available)
<code>fingerprint</code>	String (UNIQUE, indexed)	No	SHA-256 of <code>institution source_url title</code> — duplicate detection
<code>ai_summary</code>	Text	Yes	6-section AI summary from Ollama
<code>ai_topics</code>	String	Yes	Comma-separated topic slugs
<code>ai_relevance</code>	Text	Yes	BOT relevance note + disclaimer
<code>ai_processed</code>	Boolean	No	Has the AI pipeline run?
<code>ai_processed_at</code>	DateTime	Yes	When AI last ran
<code>ai_engine</code>	String	Yes	<code>ollama</code> or <code>rule-based</code>
<code>hidden</code>	Boolean (indexed)	No	Excluded from analyst views (admin can hide)
<code>manual</code>	Boolean	No	Added manually vs. collected

collected_at	DateTime	No	When the system stored it
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Design notes

- Fingerprint is UNIQUE at the DB level — even if code has a bug, the database rejects duplicates.
- AI columns are nullable — a publication is stored first, enriched second.
- hidden replaces delete — nothing is ever lost; admins can restore.
- Indexes: institution, published_date, fingerprint, hidden.

4. Table: `sources`

User-added RSS sources (the 4 built-ins live in code, not in this table).

Column	Type	Purpose
<code>id</code>	Integer	Primary key
<code>name</code>	String (UNIQUE)	Source name
<code>url</code>	String	RSS feed URL
<code>method</code>	String	Currently always <code>rss</code>
<code>active</code>	Boolean	Enabled/disabled
<code>builtin</code>	Boolean	Reserved for future use
<code>added_by</code>	String	Admin username
<code>added_at</code>	DateTime	When added

5. Table: `alert_rules`

User-defined watch rules for notifications.

Column	Type	Purpose
<code>id</code>	Integer	Primary key
<code>keyword</code>	String	Text to match (case-insensitive)
<code>source</code>	String	<code>all</code> or specific institution

topic	String	<code>all</code> or specific topic slug
priority	String	high / medium / low
created_by	String	Username
created_at	DateTime	
active	Boolean	Disable without deleting

Matching rules (all must be true)

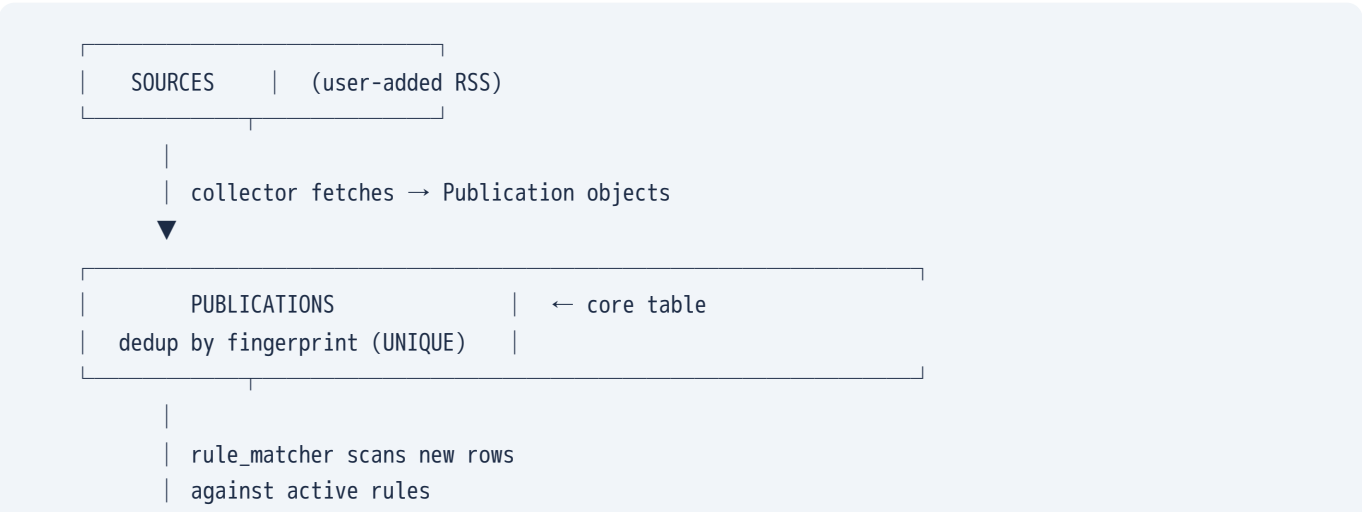
- 1. Rule source is `all` , OR equals the publication's institution
- 2. Rule topic is `all` , OR is in the publication's ai_topics
- 3. Rule keyword appears in title, abstract, ai_summary, or ai_topics

6. Table: `rule_matches`

Records every match between a rule and a publication.

Column	Type	Purpose
<code>id</code>	Integer	Primary key
<code>rule_id</code>	Integer (indexed)	References alert_rules.id
<code>publication_id</code>	Integer (indexed)	References publications.id
<code>matched_at</code>	DateTime	When the match was created

7. Data Flow





8. Users

Users are stored in memory in `backend/app/users.py` , not in the database. This is intentional for the prototype:

- No password hashing (prototype only — documented)
- Resets when the backend restarts
- Upgrade path: add a `users` table with hashed passwords

9. Known Limitations

Limitation	Impact	Fix if needed
No users table	Logins reset on restart	Add User model + hashed passwords
ai_topics is comma-separated string	Harder SQL joins for topic queries	Add publication_topics M2M table
No FK constraint on rule_matches	Orphan rows possible	Add ON DELETE CASCADE
No UNIQUE on (rule_id, publication_id)	Theoretical duplicate matches	Add UNIQUE constraint
Plain-text password (in-memory)	Security risk if compromised	Use bcrypt/argon2
No migrations	ALTER TABLE done manually	Add Alembic
SQLite write concurrency	Locks under heavy load	Switch to PostgreSQL

10. Example Queries

Count publications by source

```
SELECT institution, COUNT(*) FROM publications
WHERE hidden = 0
GROUP BY institution
ORDER BY COUNT(*) DESC;
```

Find fallback rows that need reprocessing

```
SELECT id, title FROM publications
WHERE ai_engine = 'rule-based' AND hidden = 0;
```

Recent rule matches

```
SELECT r.keyword, p.title, m.matched_at
FROM rule_matches m
JOIN alert_rules r ON r.id = m.rule_id
JOIN publications p ON p.id = m.publication_id
ORDER BY m.matched_at DESC
LIMIT 10;
```

Orphaned publications

```
SELECT institution, COUNT(*) FROM publications
WHERE institution NOT IN ('BIS', 'IMF', 'World Bank', 'CBK')
  AND institution NOT IN (SELECT name FROM sources)
GROUP BY institution;
```